

LABORATORY RECORD 2

Course :	Full Stack Web Development
Course Code:	23CM4121
Experiment No.:	2
Student Name:	R. Srivalli
Roll No.:	A24126552108

1. TITLE

Design Webpage using HTML5 & CSS3

2. AIM

To design and develop a **modern, responsive webpage using HTML5 and CSS3**, incorporating semantic HTML5 elements, CSS Flexbox, CSS Grid, gradients, transitions, transforms, styled form elements, and responsive design techniques..

3. OBJECTIVE

1. To understand the basic structure and features of **HTML5**.
 2. To apply **CSS3** properties for styling and page layout.
 3. To implement **Flexbox and CSS Grid** for arranging webpage elements.
 4. To use **gradients, transitions, transforms, and hover effects**.
 5. To design and style **HTML5 form elements**.
 6. To create a **responsive webpage** that works on desktop, tablet, and mobile devices.
 7. To develop a **modern and well-structured college webpage** using HTML5 and CSS3
-

4. THEORY

HTML5 is the standard markup language used to structure and present content on webpages. It provides semantic elements such as <header>, <nav>, <main>, <section>, <article>, and <footer> to organize webpage content clearly.

CSS3 is used to style and design HTML webpages. It provides properties for controlling colors, fonts, spacing, borders, backgrounds, and layouts. CSS3 also supports advanced features such as **Flexbox, CSS Grid, gradients, transitions, transforms, and responsive media queries**.

In this experiment, HTML5 is used to create the structure of the **ANITS Digital Campus** webpage, while CSS3 is used to provide a modern visual design and responsive layout. **Flexbox** is used for the quick-information section, **CSS Grid** is used for the academic and facility layouts, and **media queries** are used to make the webpage suitable for tablets and mobile devices.

5. ALGORITHM / PROCEDURE

1. Create a basic **HTML5 document structure** using `<!DOCTYPE html>`, `<html>`, `<head>`, and `<body>`.
 2. Add a **header and navigation bar** using semantic HTML5 elements.
 3. Create the **hero section** with heading, description, buttons, and campus image.
 4. Add the **About, Academics, Campus, and Contact** sections.
 5. Use **Flexbox** to arrange the quick-information items.
 6. Use **CSS Grid** to arrange the academic course cards and campus facilities.
 7. Apply CSS3 properties such as **colors, gradients, borders, shadows, transitions, transforms, and hover effects**.
 8. Create and style the **student enquiry form** using HTML5 form elements.
 9. Add **media queries** to make the webpage responsive on tablets and mobile devices.
 10. Link the external CSS file to the HTML document and save both files.
 11. Open the HTML file in a web browser and verify the layout, navigation, styling, form, and responsive behavior.
-

6. PROGRAM

A : Index.html

```
<!DOCTYPE html>
<html lang="en">

<head>

  <!-- Basic HTML5 document information -->
  <meta charset="UTF-8">

  <!-- Makes the webpage responsive on different devices -->
  <meta name="viewport" content="width=device-width, initial-scale=1.0">

  <title>ANITS | Digital Campus</title>

  <!-- External CSS3 stylesheet -->
  <link rel="stylesheet" href="stylerec2.css">

</head>

<body>
```

```
<!-- =====
HEADER
Demonstrates semantic HTML5 header and navigation
===== -->

<header class="main-header">

  <div class="logo-area">

    <div class="logo-box">
      ANITS
    </div>

    <div class="college-name">

      <h1>
        Anil Neerukonda Institute
      </h1>

      <p>
        of Technology and Sciences
      </p>

    </div>

  </div>

  <!-- HTML5 navigation element -->

  <nav class="navigation">

    <a href="#home">Home</a>

    <a href="#about">About</a>

    <a href="#academics">Academics</a>

    <a href="#campus">Campus</a>

    <a href="#contact">Contact</a>

  </nav>

</header>

<!-- =====
MAIN CONTENT
===== -->
```

```
<main>

<!-- =====
HERO SECTION
===== -->

<section id="home" class="hero-section">

  <div class="hero-text">

    <span class="small-label">
      ENGINEERING • TECHNOLOGY • INNOVATION
    </span>

    <h2>
      Explore the
      <span>ANITS</span>
      Campus
    </h2>

    <p>
      An academic environment designed to encourage
      technical knowledge, practical learning,
      creativity and innovation.
    </p>

    <div class="hero-buttons">

      <a href="#academics" class="primary-button">
        View Academics
      </a>

      <a href="#campus" class="secondary-button">
        Explore Campus
      </a>

    </div>

  </div>

  <!-- Campus image from observation/src -->

  <div class="hero-image">

    <div class="image-frame">

    </div>

  </div>

</section>

</main>
```

```

    <div class="image-caption">
      <strong>ANITS Campus</strong>
      <span>Sangivalasa, Visakhapatnam</span>
    </div>

  </div>

</div>

</section>

<!-- =====
QUICK INFORMATION
Demonstrates CSS Flexbox
===== -->

<section class="quick-info">

  <article class="info-item">

    <span class="info-number">01</span>

    <div>
      <h3>Technical Education</h3>
      <p>Engineering and technology focused learning.</p>
    </div>

  </article>

  <article class="info-item">

    <span class="info-number">02</span>

    <div>
      <h3>Practical Learning</h3>
      <p>Academic concepts supported by practical work.</p>
    </div>

  </article>

  <article class="info-item">

    <span class="info-number">03</span>

    <div>
      <h3>Innovation</h3>
      <p>Encouraging creativity and technical exploration.</p>
    </div>

  </article>

```

```
</section>
```

```
<!-- =====  
ABOUT SECTION  
===== -->
```

```
<section id="about" class="about-section">
```

```
<div class="section-title">
```

```
<span>01 / ABOUT</span>
```

```
<h2>
```

```
  A place to learn,
```

```
  <br>
```

```
  build and grow.
```

```
</h2>
```

```
</div>
```

```
<article class="about-content">
```

```
<p class="large-text">
```

```
  Anil Neerukonda Institute of Technology and Sciences  
  is an engineering institution located at  
  Sangivalasa, Visakhapatnam, Andhra Pradesh.
```

```
</p>
```

```
<p>
```

```
  The institute provides an academic environment  
  for engineering education, practical learning,  
  projects and professional development.
```

```
</p>
```

```
<p>
```

```
  Students can develop technical knowledge while  
  participating in laboratory activities,  
  academic projects and other learning experiences.
```

```
</p>
```

```
</article>
```

```
</section>
```

```
<!-- =====  
ACADEMICS SECTION  
Demonstrates CSS Grid  
===== -->
```

```
<section id="academics" class="academics-section">
```

```
<div class="section-heading">
```

```
<div>
```

```
<span>02 / ACADEMICS</span>
```

```
<h2>
```

```
Areas of Study
```

```
</h2>
```

```
</div>
```

```
<p>
```

```
Explore some of the major engineering  
disciplines offered at ANITS.
```

```
</p>
```

```
</div>
```

```
<!-- CSS Grid container -->
```

```
<div class="course-grid">
```

```
<article class="course-card">
```

```
<div class="course-top">
```

```
<span>01</span>
```

```
<span class="arrow">
```

```
↗
```

```
</span>
```

```
</div>
```

```
<h3>
```

```
Computer Science
```

```
</h3>
```

```
<p>
```

```
Programming, software development,  
algorithms and computer technologies.
```

</p>

</article>

<article class="course-card featured">

<div class="course-top">

02

↗

</div>

<h3>

CSE (AI & ML)

</h3>

<p>

Artificial intelligence, machine learning,
data science and intelligent applications.

</p>

</article>

<article class="course-card">

<div class="course-top">

03

↗

</div>

<h3>

Electronics & Communication

</h3>

<p>

Electronics, communication systems,
embedded systems and related technologies.

</p>

</article>

```
<article class="course-card">
```

```
  <div class="course-top">
```

```
    <span>04</span>
```

```
    <span class="arrow">  
      ↗  
    </span>
```

```
  </div>
```

```
  <h3>  
    Electrical Engineering  
  </h3>
```

```
  <p>  
    Electrical systems, power technologies  
    and engineering applications.  
  </p>
```

```
</article>
```

```
<article class="course-card">
```

```
  <div class="course-top">
```

```
    <span>05</span>
```

```
    <span class="arrow">  
      ↗  
    </span>
```

```
  </div>
```

```
  <h3>  
    Mechanical Engineering  
  </h3>
```

```
  <p>  
    Engineering design, manufacturing,  
    mechanics and mechanical systems.  
  </p>
```

```
</article>
```

```
<article class="course-card">
```

```

    <div class="course-top">

        <span>06</span>

        <span class="arrow">
            ↗
        </span>

    </div>

    <h3>
        Civil Engineering
    </h3>

    <p>
        Construction, infrastructure,
        structures and civil engineering systems.
    </p>

</article>

</div>

</section>

<!-- =====
CAMPUS SECTION
===== -->

<section id="campus" class="campus-section">

    <div class="campus-heading">

        <span>03 / CAMPUS</span>

        <h2>
            Life beyond
            <br>
            the classroom.
        </h2>

    </div>

    <div class="facility-layout">

        <article class="facility-card">

            <div class="facility-icon">
                01
            
```

```
</div>
```

```
<h3>
```

```
Laboratories
```

```
</h3>
```

```
<p>
```

```
Practical environments where students  
can apply concepts and perform experiments.
```

```
</p>
```

```
</article>
```

```
<article class="facility-card">
```

```
<div class="facility-icon">
```

```
02
```

```
</div>
```

```
<h3>
```

```
Library
```

```
</h3>
```

```
<p>
```

```
A study-oriented environment supporting  
academic and technical learning.
```

```
</p>
```

```
</article>
```

```
<article class="facility-card">
```

```
<div class="facility-icon">
```

```
03
```

```
</div>
```

```
<h3>
```

```
Classrooms
```

```
</h3>
```

```
<p>
```

```
Spaces designed for teaching, discussion  
and interaction between students and faculty.
```

```
</p>
```

```
</article>
```

```
<article class="facility-card">
```

```
<div class="facility-icon">
  04
</div>
```

```
<h3>
  Student Activities
</h3>
```

```
<p>
  Opportunities for students to participate
  in projects and extracurricular activities.
</p>
```

```
</article>
```

```
</div>
```

```
</section>
```

```
<!-- =====
CONTACT / ENQUIRY FORM
Demonstrates HTML5 form elements
===== -->
```

```
<section id="contact" class="contact-section">
```

```
<div class="contact-intro">
```

```
<span>04 / CONNECT</span>
```

```
<h2>
  Student
  <br>
  Enquiry
</h2>
```

```
<p>
  Fill in the form to submit a basic
  academic enquiry.
</p>
```

```
</div>
```

```
<form class="contact-form">
```

```
<div class="form-row">
```

```
<div class="form-field">
```

```
<label for="student-name">
  Full Name
</label>

<input
  type="text"
  id="student-name"
  name="student-name"
  placeholder="Enter your name"
  required
>

</div>

<div class="form-field">

  <label for="email">
    Email Address
  </label>

  <input
    type="email"
    id="email"
    name="email"
    placeholder="example@email.com"
    required
  >

</div>

</div>

<div class="form-row">

  <div class="form-field">

    <label for="visit-date">
      Preferred Visit Date
    </label>

    <input
      type="date"
      id="visit-date"
      name="visit-date"
    >

  </div>

  <div class="form-field">
```

```
<label for="branch">
  Interested Branch
</label>

<select id="branch" name="branch">

  <option value="">
    Select a branch
  </option>

  <option value="cse">
    Computer Science
  </option>

  <option value="aiml">
    CSE (AI & ML)
  </option>

  <option value="ece">
    Electronics & Communication
  </option>

  <option value="eee">
    Electrical Engineering
  </option>

  <option value="mech">
    Mechanical Engineering
  </option>

  <option value="civil">
    Civil Engineering
  </option>

</select>
```

```
</div>
```

```
</div>
```

```
<div class="form-field">
```

```
  <label for="message">
    Message
  </label>
```

```
  <textarea
    id="message"
    name="message"
    rows="5"
```

```

        placeholder="Write your enquiry here..."
    ></textarea>

</div>

<button type="submit" class="submit-button">

    Submit Enquiry

    <span>→</span>

</button>

</form>

</section>

</main>

<!-- =====
FOOTER
===== -->

<footer class="main-footer">

    <div class="footer-brand">

        <div class="footer-logo">
            ANITS
        </div>

        <p>
            Anil Neerukonda Institute of Technology and Sciences
        </p>

    </div>

    <div class="footer-location">

        <span>LOCATION</span>

        <p>
            Sangivalasa, Visakhapatnam,
            Andhra Pradesh
        </p>

    </div>

    <div class="footer-bottom">

```

```

    <p>
      HTML5 & CSS3 Laboratory Record
    </p>

    <p>
      © ANITS
    </p>

  </div>

</footer>

</body>

</html>

```

B : style.css

```

/* =====
ANITS DIGITAL CAMPUS
Experiment 2 - HTML5 & CSS3
===== */

/* =====
1. GLOBAL SETTINGS
===== */

* {
  margin: 0;
  padding: 0;
  box-sizing: border-box;
}

html {
  scroll-behavior: smooth;
}

body {
  font-family: Arial, Helvetica, sans-serif;
  background: #f5f8fc;
  color: #152238;
  line-height: 1.6;
}

a {
  text-decoration: none;
  color: inherit;
}

```

```

img {
  max-width: 100%;
  display: block;
}

/* =====
2. HEADER
===== */

.main-header {
  min-height: 82px;

  padding: 15px 6%;

  background: #ffffff;

  display: flex;
  align-items: center;
  justify-content: space-between;

  gap: 30px;

  border-bottom: 1px solid #e1e8f0;

  position: sticky;
  top: 0;

  z-index: 1000;
}

/* Logo and college name */

.logo-area {
  display: flex;
  align-items: center;
  gap: 12px;
}

.logo-box {
  width: 48px;
  height: 48px;

  display: flex;
  align-items: center;
  justify-content: center;

  background: #075985;

  color: white;

  font-size: 12px;

```

```

font-weight: bold;

border-radius: 10px;

box-shadow:
  0 6px 15px rgba(7, 89, 133, 0.2);
}

.college-name h1 {
  font-size: 15px;
  line-height: 1.2;
}

.college-name p {
  font-size: 12px;
  color: #66768a;
}

/* Navigation */

.navigation {
  display: flex;
  align-items: center;
  gap: 5px;

  flex-wrap: wrap;
}

.navigation a {
  padding: 8px 13px;

  font-size: 13px;
  font-weight: 600;

  color: #516174;

  border-radius: 6px;

  transition:
    background 0.25s ease,
    color 0.25s ease;
}

.navigation a:hover {
  background: #e6f4fb;
  color: #075985;
}

/* =====
3. HERO SECTION
===== */

```

```
.hero-section {
  min-height: 610px;

  padding: 80px 8%;

  display: grid;

  grid-template-columns:
    1fr 1fr;

  align-items: center;

  gap: 70px;

  /* CSS3 gradient */

  background:
    linear-gradient(
      135deg,
      #eaf7ff 0%,
      #f7fbff 55%,
      #dff3fa 100%
    );

  overflow: hidden;
}

/* Hero text */

.hero-text {
  max-width: 610px;
}

.small-label {
  display: inline-block;

  margin-bottom: 18px;

  color: #087ea4;

  font-size: 11px;
  font-weight: bold;

  letter-spacing: 2px;
}

.hero-text h2 {
  font-size: clamp(42px, 6vw, 72px);

  line-height: 1.02;
```

```
letter-spacing: -2px;

margin-bottom: 25px;
}

.hero-text h2 span {
  color: #0784a8;
}

.hero-text p {
  max-width: 520px;

  color: #52657a;

  font-size: 16px;

  margin-bottom: 30px;
}

/* Hero buttons */

.hero-buttons {
  display: flex;

  gap: 12px;

  flex-wrap: wrap;
}

.primary-button,
.secondary-button {
  padding: 12px 19px;

  border-radius: 7px;

  font-size: 13px;

  font-weight: bold;

  transition:
    transform 0.25s ease,
    box-shadow 0.25s ease;
}

.primary-button {
  background: #075985;

  color: white;
}
```

```

.secondary-button {
  background: white;

  color: #075985;

  border: 1px solid #c5dce8;
}

.primary-button:hover,
.secondary-button:hover {
  transform: translateY(-3px);

  box-shadow:
    0 8px 18px rgba(7, 89, 133, 0.16);
}

/* =====
4. HERO IMAGE
===== */

.hero-image {
  display: flex;

  justify-content: center;
}

.image-frame {
  width: min(100%, 550px);

  position: relative;

  background: white;

  padding: 10px;

  border-radius: 14px;

  box-shadow:
    0 20px 50px rgba(20, 65, 95, 0.18);

  transform: rotate(1.5deg);

  transition:
    transform 0.4s ease,
    box-shadow 0.4s ease;
}

.image-frame:hover {
  transform: rotate(0deg) translateY(-5px);

  box-shadow:

```

```

    0 25px 60px rgba(20, 65, 95, 0.22);
}

.image-frame img {
    width: 100%;

    height: 340px;

    object-fit: cover;

    border-radius: 9px;
}

.image-caption {
    padding: 12px 8px 5px;

    display: flex;

    justify-content: space-between;

    gap: 10px;

    font-size: 12px;
}

.image-caption strong {
    color: #075985;
}

.image-caption span {
    color: #718096;
}

/* =====
5. QUICK INFORMATION
Demonstrates Flexbox
===== */

.quick-info {
    padding: 0 8%;

    margin-top: -35px;

    position: relative;

    z-index: 5;

    display: flex;

    gap: 15px;
}

```

```
.info-item {  
  flex: 1;  
  
  min-height: 105px;  
  
  padding: 22px;  
  
  background: white;  
  
  display: flex;  
  
  align-items: center;  
  
  gap: 16px;  
  
  border-radius: 10px;  
  
  border: 1px solid #e1eaf1;  
  
  box-shadow:  
    0 8px 25px rgba(25, 70, 100, 0.08);  
  
  transition:  
    transform 0.3s ease;  
}  
  
.info-item:hover {  
  transform: translateY(-5px);  
}  
  
.info-number {  
  font-size: 22px;  
  
  font-weight: bold;  
  
  color: #0a91b5;  
}  
  
.info-item h3 {  
  font-size: 14px;  
  
  margin-bottom: 4px;  
}  
  
.info-item p {  
  font-size: 12px;  
  
  color: #758397;  
}
```

```
/* =====  
6. ABOUT SECTION  
===== */
```

```
.about-section {  
  padding: 120px 8% 100px;  
  
  display: grid;  
  
  grid-template-columns:  
    0.9fr 1.1fr;  
  
  gap: 80px;  
  
  background: white;  
}  
  
.section-title span,  
.campus-heading span,  
.contact-intro span {  
  color: #0784a8;  
  
  font-size: 11px;  
  
  font-weight: bold;  
  
  letter-spacing: 2px;  
}  
  
.section-title h2,  
.campus-heading h2,  
.contact-intro h2 {  
  font-size: clamp(32px, 4vw, 48px);  
  
  line-height: 1.1;  
  
  margin-top: 15px;  
  
  letter-spacing: -1px;  
}  
  
.about-content {  
  padding-top: 8px;  
}  
  
.about-content p {  
  color: #637387;  
  
  margin-bottom: 18px;  
  
  max-width: 650px;
```

```

}

.about-content .large-text {
  color: #24364c;

  font-size: 19px;

  line-height: 1.7;
}

/* =====
7. ACADEMICS SECTION
===== */

.academics-section {
  padding: 100px 8%;

  background: #f0f7fb;
}

.section-heading {
  margin-bottom: 45px;

  display: flex;

  align-items: end;

  justify-content: space-between;

  gap: 30px;
}

.section-heading span {
  color: #0784a8;

  font-size: 11px;

  font-weight: bold;

  letter-spacing: 2px;
}

.section-heading h2 {
  font-size: 42px;

  margin-top: 8px;
}

.section-heading > p {
  max-width: 380px;

```

```
color: #697b8d;

font-size: 13px;
}

/* =====
8. COURSE GRID
Demonstrates CSS Grid
===== */

.course-grid {
  display: grid;

  grid-template-columns:
    repeat(3, 1fr);

  gap: 18px;
}

.course-card {
  min-height: 240px;

  padding: 27px;

  background: white;

  border: 1px solid #dce7ef;

  border-radius: 10px;

  position: relative;

  overflow: hidden;

  transition:
    transform 0.3s ease,
    box-shadow 0.3s ease,
    border-color 0.3s ease;
}

.course-card::after {
  content: "";

  position: absolute;

  width: 90px;
  height: 90px;

  right: -35px;
  bottom: -35px;
```

```
border-radius: 50%;

background: #e5f6fb;
}

.course-card:hover {
  transform: translateY(-7px);

  border-color: #a9d9e8;

  box-shadow:
    0 15px 35px rgba(25, 70, 100, 0.11);
}

.course-card.featured {
  background:
    linear-gradient(
      145deg,
      #075985,
      #087ea4
    );

  color: white;

  border-color: transparent;
}

.course-card.featured p {
  color: #d9f3fa;
}

.course-card.featured .course-top span {
  color: #bcecf5;
}

.course-top {
  display: flex;

  justify-content: space-between;

  margin-bottom: 45px;

  font-size: 12px;

  font-weight: bold;
}

.course-top > span:first-child {
  color: #0784a8;
}
```

```

.arrow {
  font-size: 19px;
}

.course-card h3 {
  max-width: 270px;

  font-size: 21px;

  margin-bottom: 12px;
}

.course-card p {
  color: #738397;

  font-size: 13px;

  position: relative;

  z-index: 2;
}

/* =====
9. CAMPUS SECTION
===== */

.campus-section {
  padding: 110px 8%;

  background: #ffffff;
}

.campus-heading {
  margin-bottom: 45px;
}

.facility-layout {
  display: grid;

  grid-template-columns:
    repeat(4, 1fr);

  border-top: 1px solid #dce6ee;

  border-bottom: 1px solid #dce6ee;
}

.facility-card {
  min-height: 260px;

  padding: 30px 25px;

```

```

border-right: 1px solid #dce6ee;

transition:
    background 0.3s ease;
}

.facility-card:last-child {
    border-right: none;
}

.facility-card:hover {
    background: #f0f9fc;
}

.facility-icon {
    width: 42px;
    height: 42px;

    display: flex;

    align-items: center;

    justify-content: center;

    border-radius: 50%;

    background: #e4f5fa;

    color: #0784a8;

    font-size: 11px;

    font-weight: bold;

    margin-bottom: 55px;
}

.facility-card h3 {
    font-size: 19px;

    margin-bottom: 12px;
}

.facility-card p {
    color: #718094;

    font-size: 13px;
}

/* =====

```

10. CONTACT SECTION

```
===== */
```

```
.contact-section {  
  padding: 100px 8%;  
  
  display: grid;  
  
  grid-template-columns:  
    0.7fr 1.3fr;  
  
  gap: 80px;  
  
  background:  
    linear-gradient(  
      135deg,  
      #e8f7fc,  
      #f7fbfd  
    );  
}  
  
.contact-intro p {  
  margin-top: 25px;  
  
  max-width: 320px;  
  
  color: #66798d;  
}  
  
/* Form */  
  
.contact-form {  
  padding: 35px;  
  
  background: white;  
  
  border-radius: 12px;  
  
  border: 1px solid #dce8ef;  
  
  box-shadow:  
    0 15px 40px rgba(25, 75, 105, 0.08);  
}  
  
.form-row {  
  display: grid;  
  
  grid-template-columns:  
    repeat(2, 1fr);  
  
  gap: 20px;
```

```
}

.form-field {
  margin-bottom: 22px;

  display: flex;

  flex-direction: column;
}

.form-field label {
  margin-bottom: 7px;

  color: #35465a;

  font-size: 12px;

  font-weight: bold;
}

.form-field input,
.form-field select,
.form-field textarea {
  width: 100%;

  padding: 12px 13px;

  border: 1px solid #ccdbe5;

  border-radius: 6px;

  background: #fbfdfe;

  color: #26384c;

  font-family: inherit;

  font-size: 13px;

  outline: none;

  transition:
    border-color 0.25s ease,
    box-shadow 0.25s ease;
}

.form-field textarea {
  resize: vertical;
}

.form-field input:focus,
```

```
.form-field select:focus,
.form-field textarea:focus {
  border-color: #0784a8;

  box-shadow:
    0 0 0 3px rgba(7, 132, 168, 0.1);
}

/* Submit button */

.submit-button {
  width: 100%;

  padding: 13px 18px;

  border: none;

  border-radius: 7px;

  background: #075985;

  color: white;

  font-family: inherit;

  font-size: 13px;

  font-weight: bold;

  cursor: pointer;

  display: flex;

  justify-content: space-between;

  align-items: center;

  transition:
    background 0.25s ease,
    transform 0.25s ease;
}

.submit-button span {
  font-size: 20px;
}

.submit-button:hover {
  background: #087ea4;

  transform: translateY(-2px);
}
```

```
/* =====  
11. FOOTER  
===== */  
  
.main-footer {  
  padding: 55px 8% 25px;  
  
  background: #102a43;  
  
  color: white;  
}  
  
.footer-brand {  
  display: flex;  
  
  align-items: center;  
  
  gap: 15px;  
  
  margin-bottom: 45px;  
}  
  
.footer-logo {  
  width: 48px;  
  height: 48px;  
  
  display: flex;  
  
  justify-content: center;  
  align-items: center;  
  
  border-radius: 9px;  
  
  background: #22a6c3;  
  
  font-size: 11px;  
  
  font-weight: bold;  
}  
  
.footer-brand p {  
  color: #b7c9d8;  
  
  font-size: 13px;  
}  
  
.footer-location {  
  margin-bottom: 35px;  
}
```

```

.footer-location span {
  color: #65d0e5;

  font-size: 10px;

  letter-spacing: 2px;

  font-weight: bold;
}

.footer-location p {
  margin-top: 7px;

  color: #d2dfe9;

  font-size: 13px;
}

.footer-bottom {
  padding-top: 20px;

  border-top: 1px solid #345069;

  display: flex;

  justify-content: space-between;

  gap: 20px;

  color: #91a8bb;

  font-size: 11px;
}

```

```

/* =====
12. TABLET RESPONSIVE DESIGN
===== */

```

```

@media (max-width: 950px) {

  .main-header {
    flex-direction: column;

    align-items: flex-start;
  }

  .navigation {
    width: 100%;
  }

  .hero-section {

```

```

    grid-template-columns: 1fr;

    gap: 45px;

    padding-top: 65px;
}

.hero-image {
    justify-content: flex-start;
}

.course-grid {
    grid-template-columns:
        repeat(2, 1fr);
}

.facility-layout {
    grid-template-columns:
        repeat(2, 1fr);
}

.facility-card:nth-child(2) {
    border-right: none;
}

.facility-card:nth-child(1),
.facility-card:nth-child(2) {
    border-bottom: 1px solid #dce6ee;
}

.about-section,
.contact-section {
    grid-template-columns: 1fr;

    gap: 40px;
}
}

/* =====
13. MOBILE RESPONSIVE DESIGN
===== */

@media (max-width: 650px) {

    .main-header {
        padding: 15px 5%;
    }

    .college-name h1 {
        font-size: 13px;

```

```
}

.college-name p {
  font-size: 10px;
}

.navigation {
  gap: 2px;
}

.navigation a {
  padding: 6px 8px;

  font-size: 11px;
}

.hero-section {
  padding: 55px 5%;
}

.hero-text h2 {
  font-size: 45px;

  letter-spacing: -1px;
}

.hero-text p {
  font-size: 14px;
}

.image-frame img {
  height: 240px;
}

.image-caption {
  flex-direction: column;
}

.quick-info {
  padding: 0 5%;

  margin-top: 20px;

  flex-direction: column;
}

.about-section,
.academics-section,
.campus-section,
.contact-section {
  padding-left: 5%;
}
```

```
padding-right: 5%;
}

.about-section {
padding-top: 75px;
}

.section-heading {
flex-direction: column;

align-items: flex-start;
}

.section-heading h2 {
font-size: 34px;
}

.course-grid {
grid-template-columns: 1fr;
}

.facility-layout {
grid-template-columns: 1fr;
}

.facility-card,
.facility-card:nth-child(2) {
border-right: none;

border-bottom: 1px solid #dce6ee;
}

.facility-card:last-child {
border-bottom: none;
}

.form-row {
grid-template-columns: 1fr;

gap: 0;
}

.contact-form {
padding: 23px;
}

.footer-brand {
align-items: flex-start;

flex-direction: column;
}
```

```

.footer-bottom {
    flex-direction: column;
}
}

```

7. CODE EXPLANATION

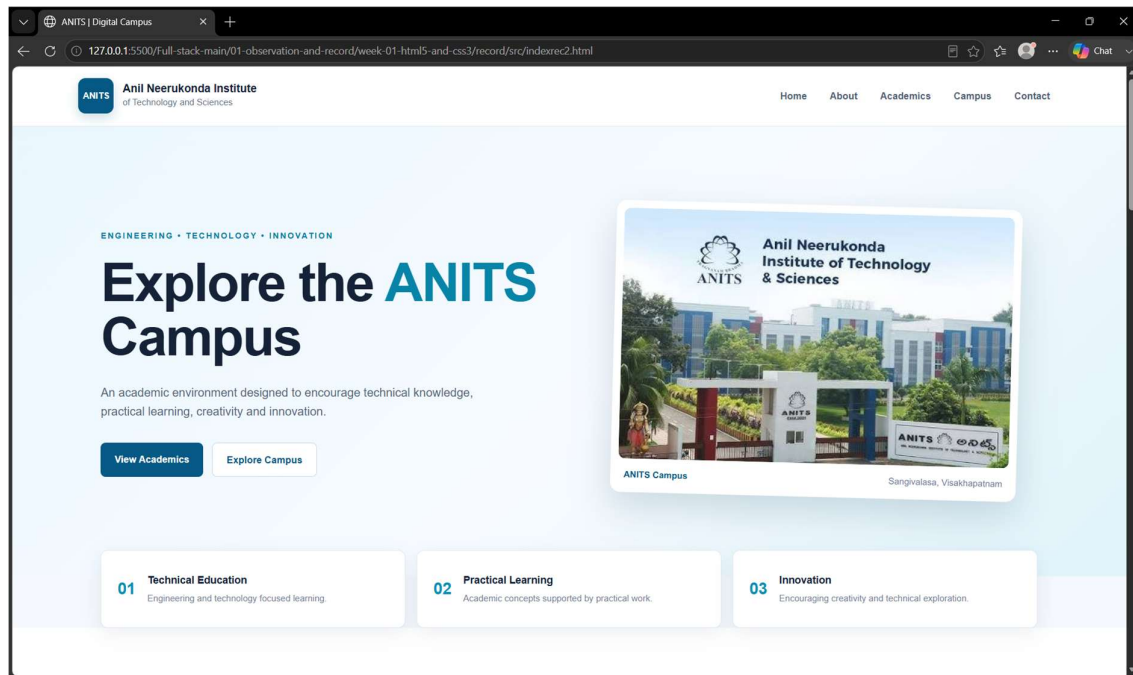
S.No.	HTML/CSS Component	Explanation
1	<!DOCTYPE html>	Declares the document as an HTML5 webpage.
2	<meta>	Defines character encoding and responsive viewport settings.
3	<header>	Creates the main header containing the ANITS logo and college name.
4	<nav>	Provides navigation links to different sections of the webpage.
5	<main>	Contains the main content of the webpage.
6	<section>	Divides the webpage into Home, About, Academics, Campus, and Contact sections.
7	<article>	Represents independent content blocks such as information and course cards.
8		Displays the ANITS campus image.
9	<form>	Creates the student enquiry form.
10	<input>	Accepts user information such as name, email, and date.
11	<select>	Provides a dropdown list for selecting a branch.
12	<textarea>	Allows the user to enter a detailed message.
13	<footer>	Displays location and laboratory record information.
14	display: flex	Arranges elements using the CSS Flexbox layout.
15	display: grid	Creates structured layouts using CSS Grid.
16	linear-gradient()	Applies gradient backgrounds to selected sections.
17	transition	Creates smooth changes during hover and focus effects.
18	transform	Adds visual movement and rotation effects to elements.

19	box-shadow	Adds depth and visual emphasis to cards and containers.
20	@media	Implements responsive design for tablets and mobile devices.
21	:hover	Applies styling changes when the user moves the pointer over an element.
22	clamp()	Provides flexible font sizing according to the screen size.

8. TEST CASES AND EXECUTION DATA

Test Case 1 – Webpage Loading

Input	Action	Expected Result	Status
indexrec2.html	Open the HTML file in a web browser	ANITS Digital Campus webpage loads successfully with header, navigation, hero section, and campus image.	Pass



Test Case 2 – Navigation Links

Input	Action	Expected Result	Status
Home, About, Academics, Campus, Contact	Click each navigation link	The webpage smoothly navigates to the corresponding section.	Pass

01 / ABOUT

A place to learn, build and grow.

Anil Neerukonda Institute of Technology and Sciences is an engineering institution located at Sangivalasa, Visakhapatnam, Andhra Pradesh.

The institute provides an academic environment for engineering education, practical learning, projects and professional development.

Students can develop technical knowledge while participating in laboratory activities, academic projects and other learning experiences.

Test Case 3 – Hero Section Buttons

Input	Action	Expected Result	Status
View Academics / Explore Campus	Click the respective button	The webpage moves to the Academics or Campus section.	Pass

02 / ACADEMICS

Areas of Study

Explore some of the major engineering disciplines offered at ANITS.

01

Computer Science

Programming, software development, algorithms and computer technologies.

02

CSE (AI & ML)

Artificial intelligence, machine learning, data science and intelligent applications.

03

Electronics & Communication

Electronics, communication systems, embedded systems and related technologies.

04

Electrical Engineering

Electrical systems, power technologies and engineering applications.

05

Mechanical Engineering

Engineering design, manufacturing, mechanics and mechanical systems.

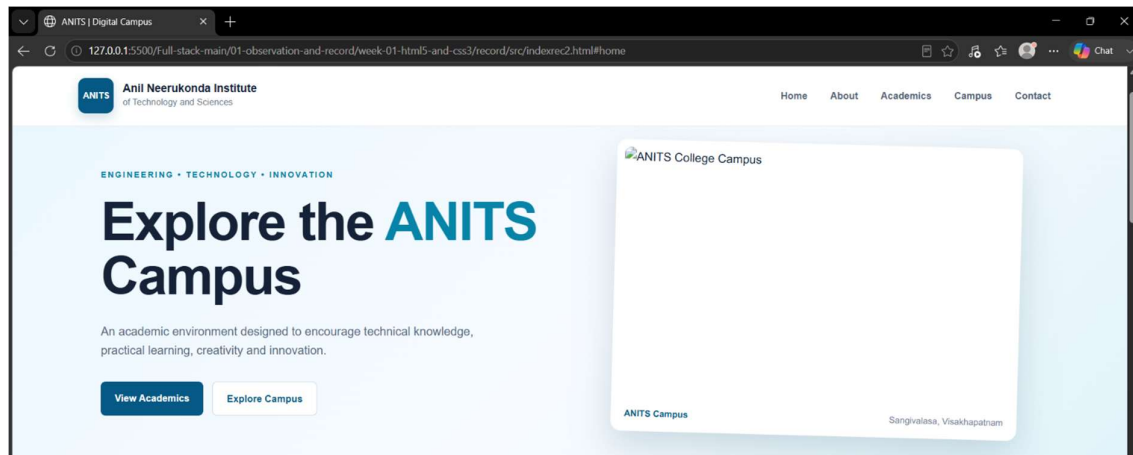
06

Civil Engineering

Construction, infrastructure, structures and civil engineering systems.

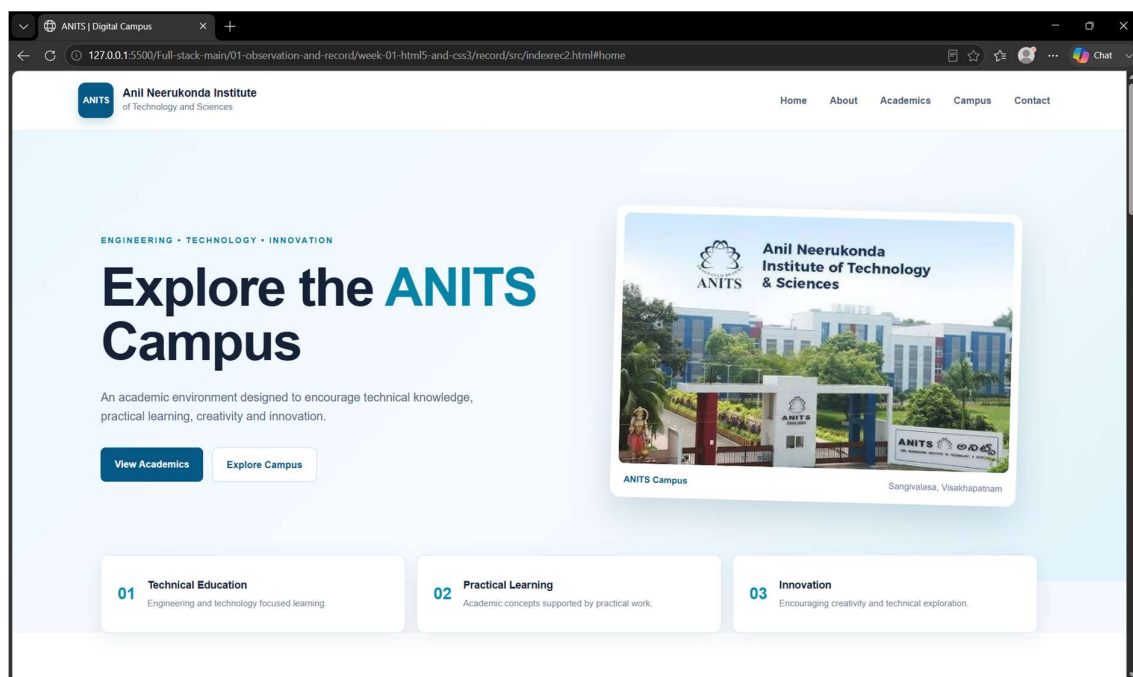
Test Case 4 – Missing Image

Input	Action	Expected Result	Status
Incorrect campus.jpg path	Reload the webpage	Campus image fails to load and alternative text is displayed.	Fail

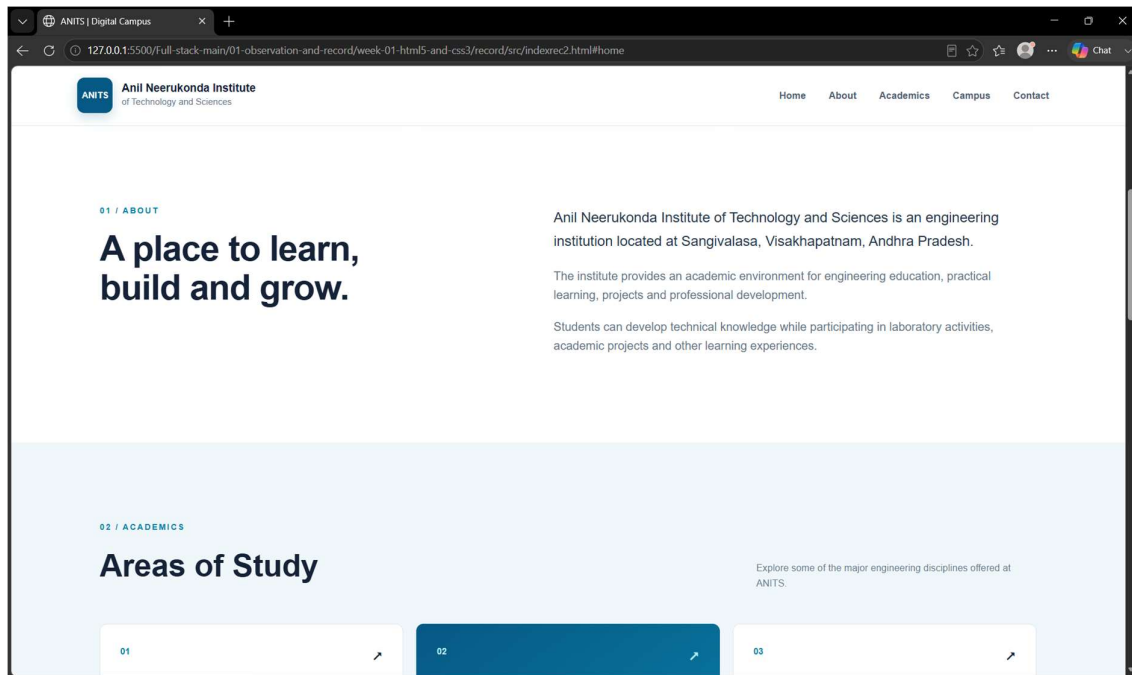


9. OUTPUT

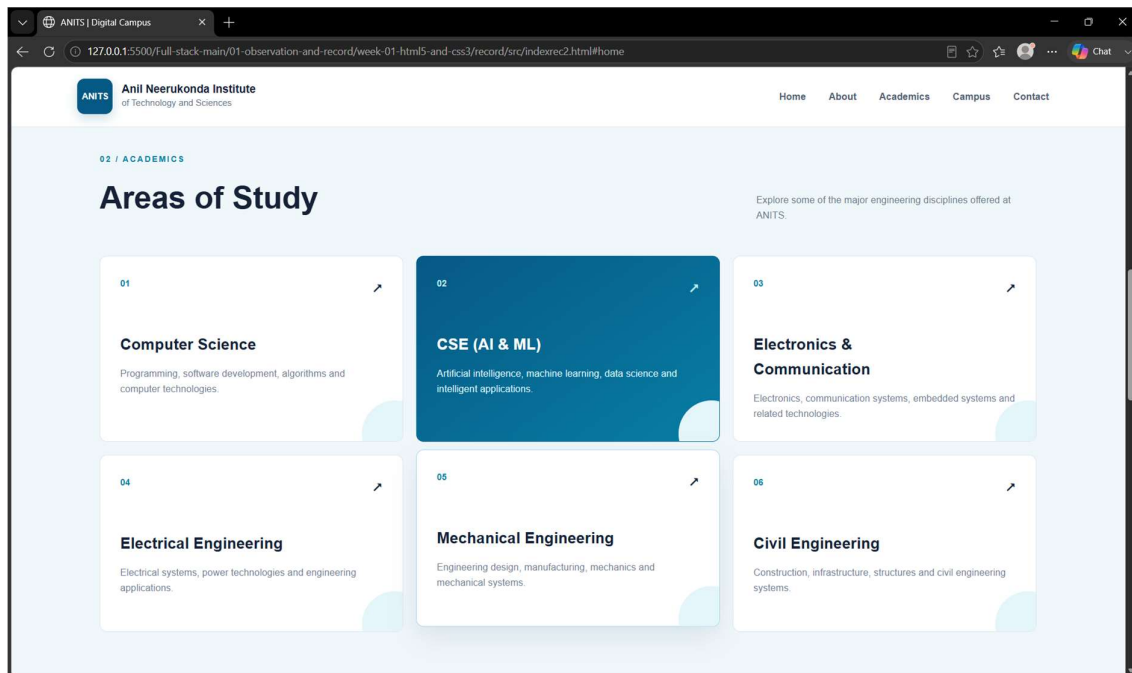
Output 1:



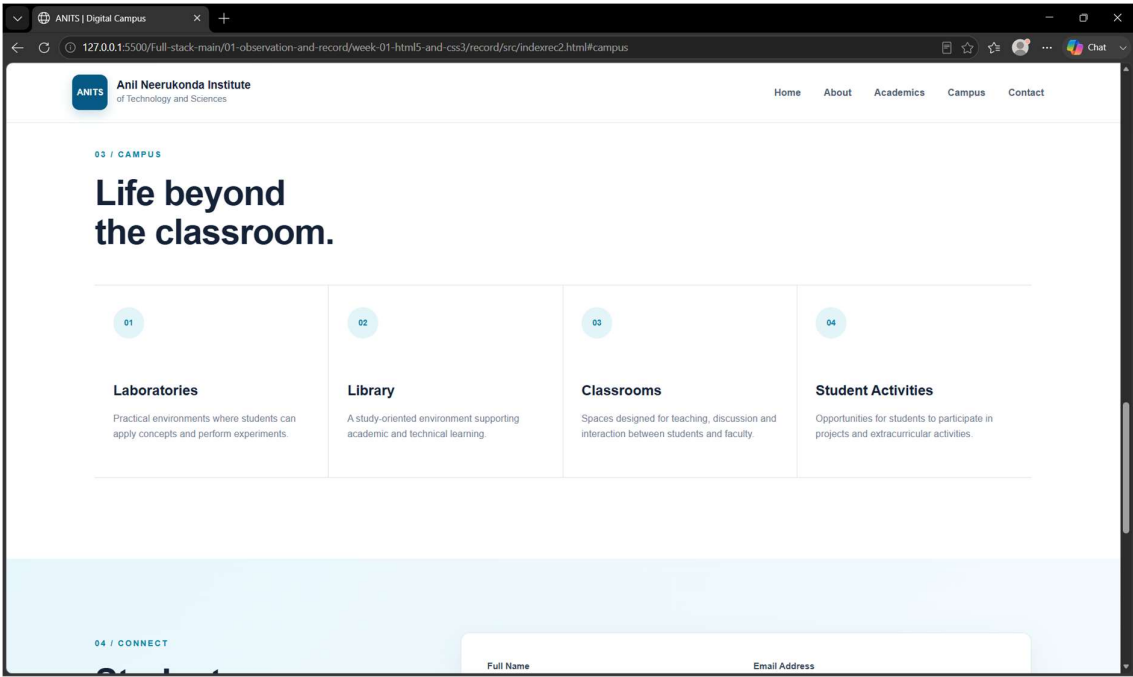
Output 2:



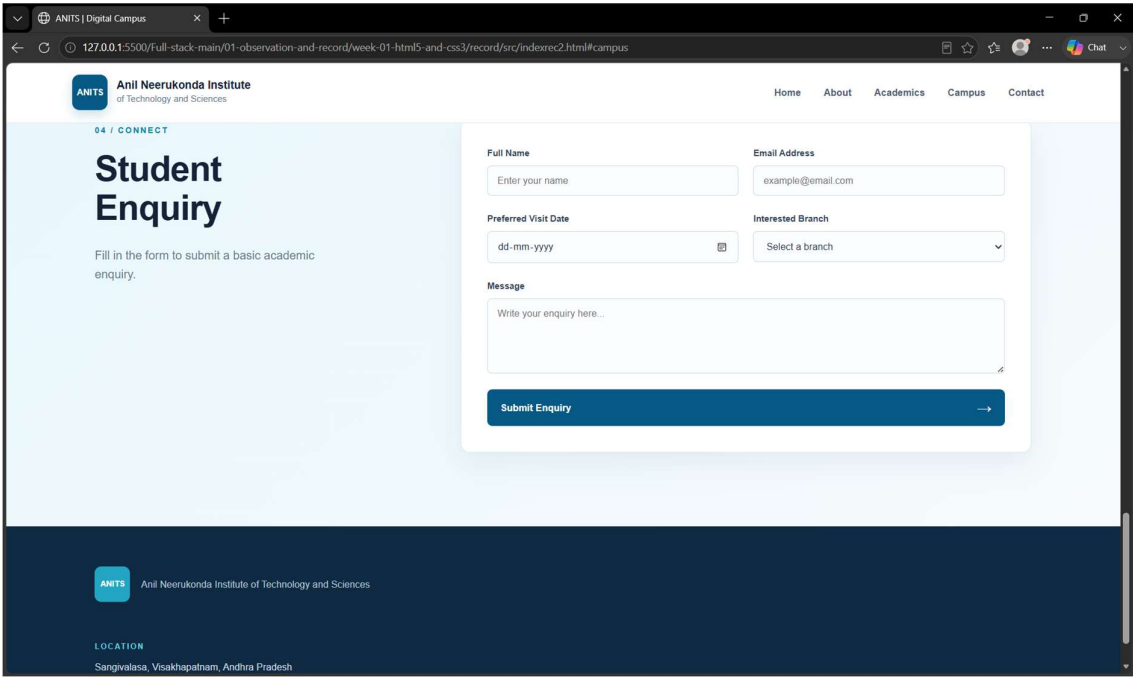
Output 3:



Output 4:



Output 5:



10. RESULT

The **webpage** was successfully designed using **HTML5 and CSS3**, implementing responsive layouts, Flexbox, CSS Grid, styling, and form elements.